

01 MOTIVATION

The integration gap

Public signals—GDP, jobs, wages, taxes, competition—sit across five sources with different formats and vintages.

Our system

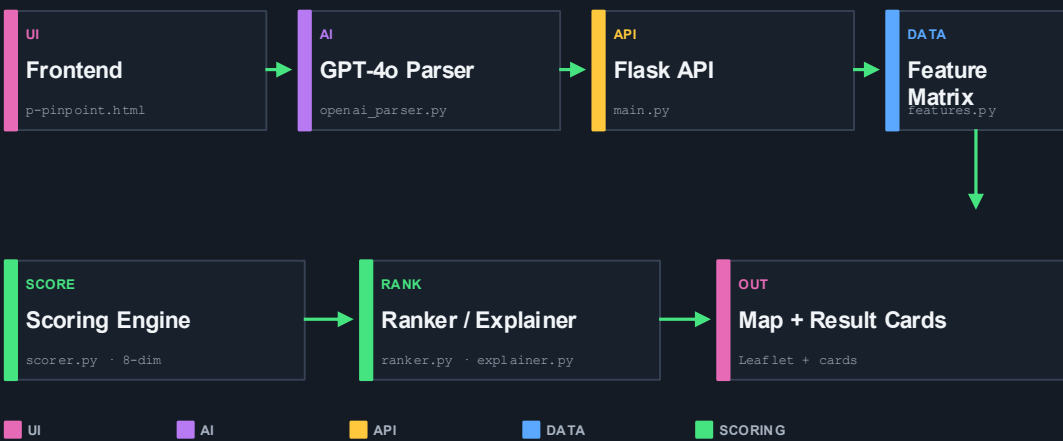
Chat input → GPT-4o profile → county feature matrix → 8-dimension weighted score → ranked county explanations.

02 DATA SOURCES

Dataset	Scope	Signal
BEA CAGDP9	3,000 counties 2001–24	GDP × 23 ind. 5-yr CAGR
Census LODES	All counties 2021	Daytime jobs wage tiers
Census CBP	All counties 2023	Establishments / NAICS
Tax Foundation	50 + DC 2024	Top corp. income rate
BLS OEWS	50 states 2024	Median wage by occupation

Joined on 5-digit county FIPS · cached once per server boot

03 SYSTEM ARCHITECTURE



04 8-DIMENSION SCORING ENGINE

01

Industry Health

BEA

GDP share · LQ · CAGR

02

Economic Growth

BEA

5-yr CAGR · recovery

03

Market Size

BEA

Log-norm. county GDP

04

Complementary

BEA

Adjacent sector presence

05

Stability

BEA

HHI diversity · volatility

06

Competitive Density

CBP

Establishments / 1K jobs

07

Talent Supply

LODES + OEWS

High-wage share × wage cost

08

Tax Climate

Tax Found.

Corporate rate (inverted)

05 PRIORITY WEIGHTS

	Industry	Growth	Market	Complem.	Stability	Comp. Density	Talent	Tax
default	.25	.20						
foot traffic			.28					
affordable								.35
growth pot.	.18	.38				.28		
low comp.			.18					
stability					.30			
target cust.	.18		.22	.25				
lower weight ← → higher weight								
blank = low/zero weight								

ENGINEERING CONTRIBUTION

what I built end-to-end

- 01

Data Pipeline

Unified 5 public datasets into a county feature matrix.
- 02

Feature Eng. + Scoring

Designed county features and 8-dim weighted scoring.
- 03

API + Frontend

Implemented Flask API + Leaflet dashboard with GPT-4o intake.

06 USER INTERFACE

CHAT INTAKE

"I want to open a tech startup in the Pacific Northwest."
→ GPT-4o JSON profile

LEAFLET MAP

→ county pins ranked by score

RESULT CARD

King County, WA · Score XX.X
→ breakdown + explanation

07 LIMITATIONS · FUTURE

- Now

hand-tuned weights · local Flask only
- Vintages

LODES '21 · CBP '23 · BEA '24
- Next

learning-to-rank · SHAP · refresh pipeline · deployment prep